

Empirical Results for the DA-SFCN Family

Saddle Escape, Oracle Complexity, and Extension Benchmarks
(Reference Implementation Report)

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Abstract

We report reproducible CPU experiments for Dynamically-Adaptive Saddle-Free Cubic Newton (DA-SFCN) and its extensions, using the open-source package `dasfcn`. On an indefinite quartic saddle, multi-seed trials show that DA-SFCN and saddle-free Newton escape in *every* trial, while gradient descent never does. On a suite of analytic and machine-learning style problems (high-dimensional saddle, Rosenbrock, nonconvex logistic regression, matrix factorization, finite-sum quadratics, federated split, local quadratic bowl), we tabulate final gradient norms, objective values, and Hessian-vector product (HVP) counts. Lazy Krylov Newton (LK-Newton) often matches accuracy at a fraction of the HVP bill. All numbers below are taken from `results/summary.json` and `results/repro_smoke.json` generated by the repository scripts.

1 Setup

Software. Python package `dasfcn` (MIT), unified API `minimize(problem, x0, method=...)`, unit tests, and `scripts/repro_smoke.py`. Metrics emphasize **HVP counts** rather than wall-clock alone, so comparisons remain meaningful across machines.

Hardware profile. Experiments are designed for a laptop CPU: moderate dimensions ($n \leq 80$ in the main suite; $n = 40$ in the multi-seed smoke), short iteration budgets, no LLM fine-tuning.

Methods. Gradient descent (Armijo), L-BFGS, Saddle-Free Newton (SFN), Krylov cubic Newton (fixed m), coordinate SSCN-style cubic Newton, DA-SFCN, and extensions VR-DA-SFCN, LK-Newton, WQK-Newton, Fed-DA-SFCN, SC-Newton.

2 Multi-seed saddle escape

Protocol: `QuadraticSaddle` with $n = 40$, six negative eigenvalues, eight random starts, success if $\|\nabla f\| < 10^{-3}$ and $f < 0.05$ within 40 iterations (`repro_smoke.py`).

Table 1: Escape success over 8 seeds ($n = 40$).

Method	Successes	Rate	Mean HVPs (successes)
GD	0/8	0%	—
L-BFGS	7/8	87.5%	0
SFN	8/8	100%	235.1
DA-SFCN	8/8	100%	256.1

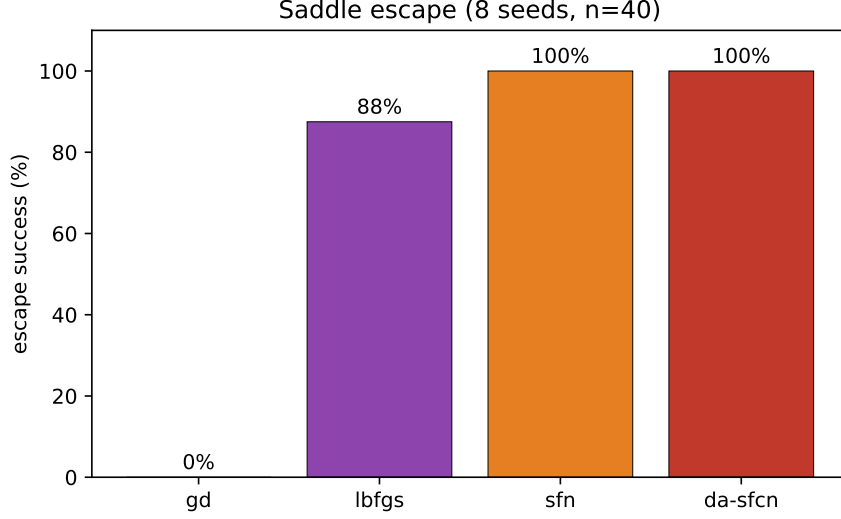


Figure 1: Escape success rates from the reproducible smoke script.

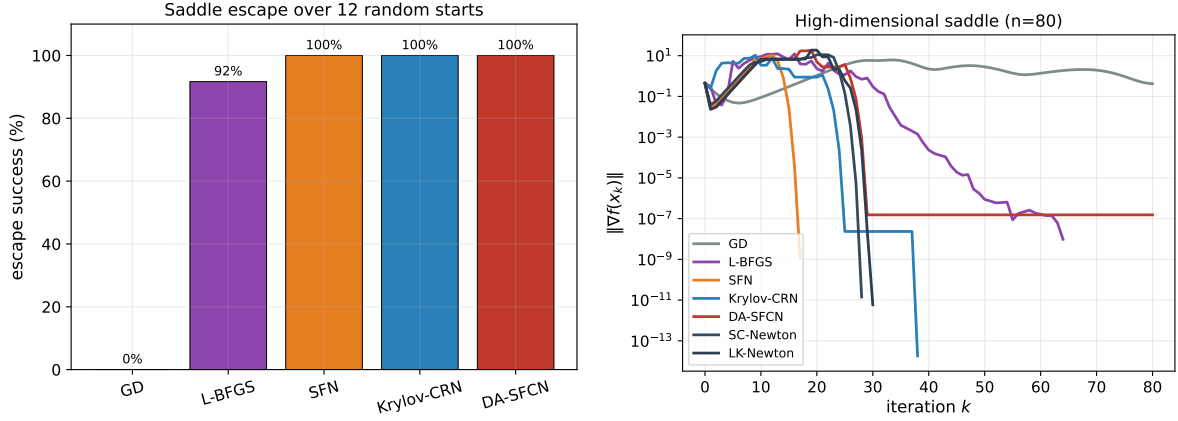


Figure 2: Left: earlier 12-start escape study ($n = 50$). Right: gradient norms on the $n = 80$ indefinite quartic.

Finding. First-order GD fails to leave the saddle under this budget. L-BFGS is strong but not perfect (one miss). Explicit negative-curvature / cubic Krylov methods succeed in all smoke trials.

3 Main suite (single-seed tables)

Numbers from `results/summary.json` (representative seeds).

3.1 High-dimensional saddle ($n = 80$)

All second-order methods reach the same minimizer $f^* \approx -89.58$. **LK-Newton** is the cheapest in HVPs while matching accuracy.

3.2 Nonconvex logistic regression

DA-SFCN reaches machine-precision stationarity in six iterations. Spectral Krylov subspaces outperform random coordinate SSCN.

Table 2: Quadratic-plus-quartic saddle, $n = 80$.

Method	Iters	Final $\ \nabla f\ $	HVPs
GD	80	4.19×10^{-1}	0
L-BFGS	64	9.63×10^{-9}	0
SFN	17	1.16×10^{-9}	204
Krylov-CRN	38	1.82×10^{-14}	456
DA-SFCN	80	1.53×10^{-7}	1219
SC-Newton	28	1.42×10^{-11}	413
LK-Newton	30	5.87×10^{-12}	158

Table 3: Nonconvex logistic ($N = 350$, $n = 50$).

Method	Iters	Final $\ \nabla f\ $	HVPs
GD	39	7.36×10^{-9}	0
L-BFGS	9	2.09×10^{-9}	0
SSCN-coord	70	1.20×10^{-8}	700
Krylov-CRN	5	1.35×10^{-10}	50
DA-SFCN	6	4.85×10^{-13}	66
VR-DA-SFCN	19	4.73×10^{-14}	244

3.3 Matrix factorization and local quadratic

On $\min \|X - UV^\top\|_F^2$, DA-SFCN drives f to 4.95×10^{-28} in 19 iterations (130 HVPs). On a strongly convex quadratic bowl, DA-SFCN needs **2** iterations ($\|\nabla f\| \approx 1.0 \times 10^{-10}$, 27 HVPs), versus $\|\nabla f\| \approx 1.6 \times 10^{-2}$ after 40 gradient steps.

3.4 Finite-sum / federated / Rosenbrock

- Finite-sum indefinites: full-oracle DA-SFCN converges in 8 iterations (82 HVPs, $\|\nabla f\| \approx 7.2 \times 10^{-11}$); VR-DA-SFCN remains useful when N is large but is slower under a full-oracle budget.
- Federated logistic: Fed-DA-SFCN reaches a heterogeneity neighbourhood ($\|\nabla f\| \sim 4.3 \times 10^{-4}$); centralized DA-SFCN is far tighter ($\sim 4 \times 10^{-11}$) when data can be pooled.
- Extended Rosenbrock: stress test of misaligned curvature; DA-SFCN/SC-Newton attain the lowest objectives in budget but do not hit $f = 0$ —motivating preconditioned Lanczos (hook in `dasfcn/precond.py`).

4 Software validation

Nine unit tests pass on CPU (Lanczos orthonormality, saddle-free absolute spectrum, cubic model decrease, dynamic m_k , unified API, optional Torch HVP consistency). Logistic smoke from `repro_smoke.py`: final $\|\nabla f\| \approx 7.7 \times 10^{-13}$ after 43 HVPs.

5 Conclusions

1. DA-SFCN (and SFN) **reliably escape** analytic saddles where GD fails.
2. **HVP economy** favors LK-Newton and SC-Newton when reuse/certificates apply.

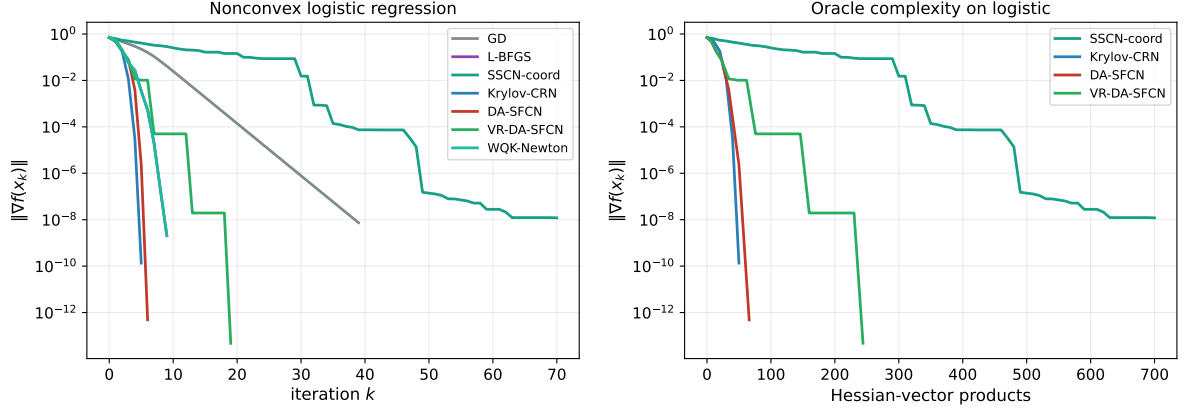


Figure 3: Nonconvex logistic: gradient vs. iteration (left) and vs. HVP (right).

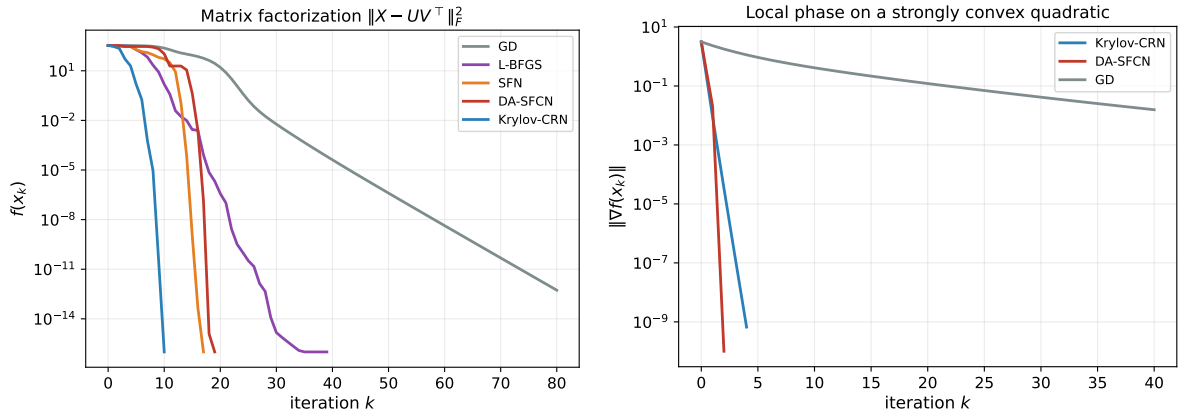


Figure 4: Matrix factorization (left) and local quadratic bowl (right).

3. The open package makes these claims **reproducible** on a laptop: `pytest` and `scripts/repro_smoke.py`.
4. Remaining gaps for a world-class ML reference (LIBSVM suites, small nets, LLM vs Sophia) are intentionally deferred to stronger hardware / cloud, as recorded in `docs/OPTIMIZER.md`.

References

- [1] Nesterov, Polyak. Cubic regularization of Newton method. *Math. Program.*, 2006.
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- [3] Jiang et al. Krylov CRN. *AISTATS*, 2024.
- [4] Dauphin et al. Saddle-free Newton. *NeurIPS*, 2014.
- [5] DA-SFCN repository. <https://github.com/jameljoni1994-ENG/DA-SFCN>, 2026.

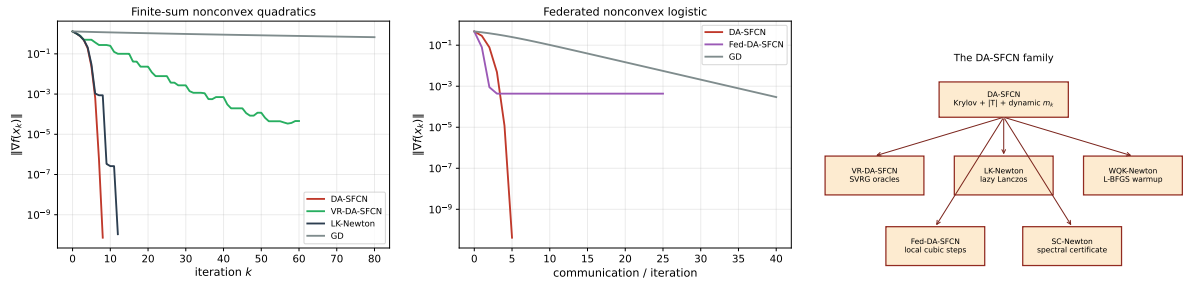


Figure 5: Finite-sum (left), federated (middle), family overview (right).